

Government Engineering College, Bharuch

Mechanical Engineering Department

Date -29/8/26

SYLLABUS FOR PROGRESSIVE ASSESSMENT EXAMINATION

B.E. -3 Year-2026-27 Semester-5

Subject Name and Code: HEAT TRANSFER - BE05000291

Date of Examination : 05/9/2026

Sr. No.	CONTENT	CO
1	Fourier's law, effect of temperature on thermal conductivity of different solids, liquids and gases, generalized equation in Cartesian, cylindrical and spherical coordinates and its reduction to specific cases, One-dimensional steady state conduction, heat conduction through plane and composite walls, cylinders and spheres, electrical analogy, critical radius of insulation for cylinder and sphere, overall heat transfer coefficient Heat transfer from extended surface: Types of fin, heat flow through uniform cross sectional area fin for various cases like infinitely long fin, fin insulated at the tip and fin losing heat at the tip, efficiency and effectiveness of fin	CO-1
3	Convection: Newton's law of cooling, Dimensional analysis applied to forced and free convection, dimensionless numbers and their physical significance, empirical correlations for free and forced convection Continuity, momentum and energy equations, thermal and hydrodynamic boundary layer	CO-2
4	Radiation: Absorptivity, reflectivity and transmissivity, black, white and grey body, emissive power, emissivity, Kirchhoff's law, Planck's law, Rayleigh-Jeans' law, Wien's law, Wien's displacement law, Stefan-Boltzmann law, intensity of radiation, radiation heat exchange between black bodies, shape factor.	CO-3

Date: 29/8/26
Place: Bharuch

Prof. D.A.Pandey
Course Coordinator

